

Safety First — A Systematic Literature Review on the Impact of AI on High-Stakes Decision-Making

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Abstract

Abstract text.

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1. Introduction

2. Methods

This analysis follows the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) methodology and therefore collects documents in a systematic and automated way. To do so, we start by defining the research questions this review seeks to answer. These include, the titular: *What is the impact of AI on high-stakes decision making?* as well as: *How are safety outcomes perturbed by AI-based decision-making in sensitive industries?* and *What are the safety risks of using AI to make decisions in sensitive industries?* While there are obvious benefits and harms to using AI-based technologies, that comparison alone is far too high-level to inspire targeted change within the AI technologies themselves. The nature of the questions guiding this review probe at what unique features and outcomes set AI-based decision-making mechanisms apart from traditional methods. Understanding these phenomena in detail would allow useful insight for both governmental agencies and engineers to mitigate evident shortcomings of this high-impact technology.



Figure 1: Directed acyclic graph showing automatic search workflow.

2.1. Query Term Compilation

After determining our research questions, we define a set of key terms using the Population, Intervention, Comparison, Outcome (PICO) methodology, and use the Cartesian product of these terms (one from each category) to determine relevant search queries. We use APIs for each of two databases, Scopus and Web of Science, to make these queries via a Python workflow, storing all resultant papers. Then, we remove duplicates based on digital object identifier (DOI), and merge the results of both database searches into a combined bibliography. Figure ?? shows visual representation of this workflow.

Critical to actually making these queries is defining the lists of keywords that comprise them. We start by defining our population terms.

We can organize the population terms we decided to include in this search in five major groups: nuclear and radiation, aviation, autonomous driving, energy and industrial processes, and medical technology and pharmaceuticals. These groups were selected for their highly sensitive nature, their propensity to use AI, and their regulation. This includes oversight by the U.S. Nuclear Regulatory Commission (NRC), the Federal Aviation Administration (FAA), the National Highway Traffic Safety Administration (NHTSA), the Occupational Safety and Health Administration (OSHA), and the Food and Drug Administration (FDA). Obviously, there are a vast number of industries

that might be considered “sensitive” and are federally regulated in the United States, however, due to time and space limitations, we prioritize those listed above in our analysis. Specifically, our goal was to avoid cases that consider risks not related to human health and safety (financial risks); we will discuss this further when we consider our exclusion criteria. The population search terms we include in our querying are shown in Table ??.

After determining our population terms, we then must consider our intervention terms. We determined this group by compiling a list of AI-related terms that would likely appear in any AI-based decision-making model. This includes words and phrases like “machine learning,” “neural networks,” and “language models.” While this set surely does not contain every possible AI-related term, its goal is to target the major ones. We presume that almost any paper using AI will use the term “AI” itself, so the goal of additional terms is mostly out of precaution. In our full query, which we describe later in this section, we use the search operator W/5 and NEAR/5 for Scopus and Web of Science, respectively, to pair each intervention term in Table ?? with the wildcard term “decision*”. In lieu of a mere AND operator, these operators require the word “decision” and its derivatives to be within five words of the AI-intervention term. The objective of this pairing is to return articles that use AI to either make or inform some kind of decision. The criticality and sensitivity of that decision will be evaluated during later screening. The intervention search terms we include in our querying are shown in Table ??.

Finally, we consider our outcome terms. Our research question focuses on the impact of AI-based decision-making on safety outcomes in sensitive industries. Thus, our outcome terms focus on safety, specifically, on risks to human health. We include general terms like “incident,” “accident,” and “near miss,” as well as more industry-specific terms, like “misdiagnos*” and “operating experience,” which cover medicine and power systems, respectively. To make this list extra comprehensive, and to ensure we capture any failures that led to official infractions, we included several regulation-specific terms too. This includes phrases like “FDA recall*” and “OSHA citation.” A complete list of outcome terms used in our querying are shown in Table ??.

2.2. Search Process

The goal of our search process was to search two databases, Scopus and Web of Science, for all articles containing our four primary search terms in their title, abstract, or keywords. We imposed two additional requirements, that the papers be published after and not including 2014 and that